



If-Else Statement in Python

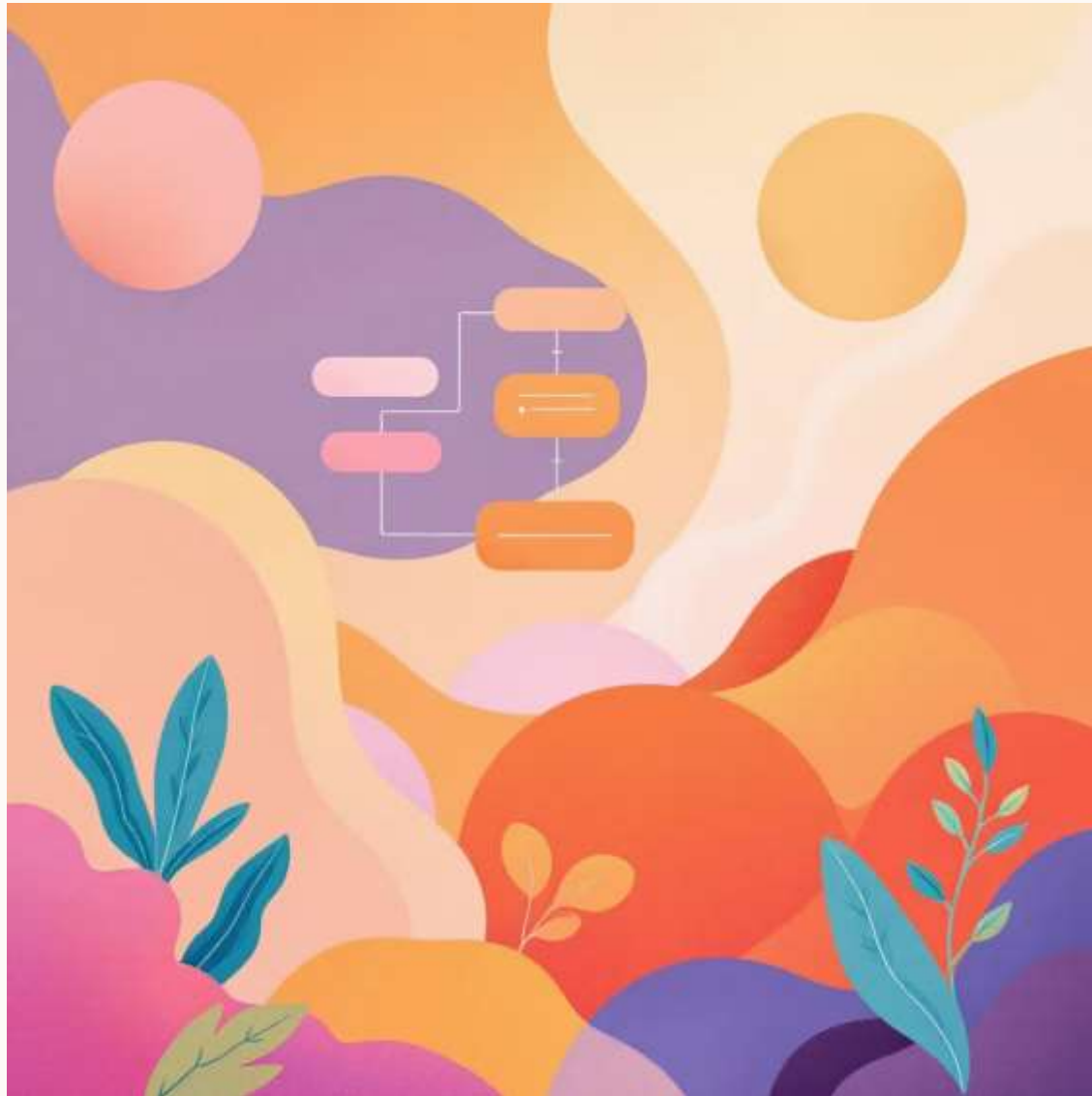
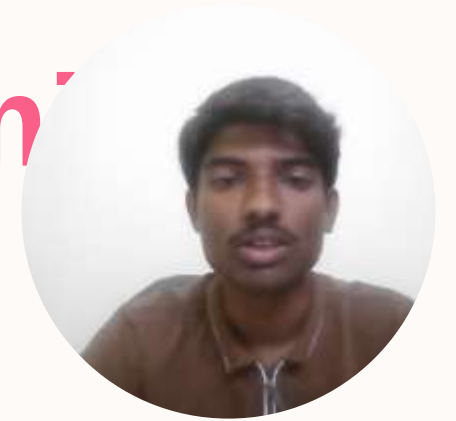
Python Basics

Name: Saiprasad Ashokrao Kawdikar

Course: EDS



What is Decision-Making in Programming?



Python is a powerful programming language that allows programs to make decisions based on specific conditions. Just like we make choices in daily life, programs use **decision-making statements** to control the flow of execution.

These statements enable programs to execute different blocks of code depending on whether certain conditions are true or false.



Understanding If–Else Statements



Conditional Logic

Checks if a condition is true or false



Two Paths

Executes one block if true, another if false



Smart Programs

Makes programs respond dynamically to data

An If–Else statement is a fundamental control structure that allows your program to choose between two different actions based on a condition. It's like a fork in the road—your program takes one path or the other depending on the condition's outcome.

Python Syntax Structure



```
if condition:
    statement1
else:
    statement2
```

01

if keyword

Begins the conditional statement

02

Condition

A logical expression that evaluates to True or False

03

Colon :

Marks the end of the condition

04

Indentation

Code blocks must be properly indented (4 spaces)

05

else keyword

Optional—executes when condition is False

Example: Student Grade Checker



Python Code

```
marks = 75

if marks >= 50:
    print("Pass!")
else:
    print("Fail!")
```

Output: Pass!

How It Works

- Variable `marks` stores the student's score
- Condition checks if marks are 50 or higher
- If True: prints "Pass!"
- If False: prints "Fail!"

📄 **Key Point:** Since `75 >= 50` is True, the program executes the first block.

Real-Life Example



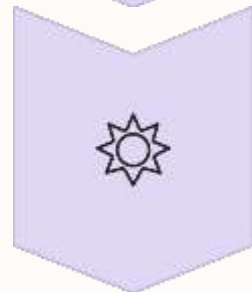
Condition

Is it raining?



True Path

Take an umbrella



False Path

Don't take an umbrella

This everyday decision mirrors how If-Else works in code. The program checks a condition (is it raining?) and takes different actions based on the answer—just like you would in real life!



Key Takeaways

Decision-Making Power

If-Else statements enable programs to make logical decisions and respond to different situations

Foundation Concept

Understanding If-Else is essential for writing more complex programs and algorithms

Real-World Application

This logic appears everywhere—from simple games to complex software systems





Common Mistakes to Avoid

✗ Missing Colon

Always include a colon `:` after the condition

✗ Wrong Indentation

Use consistent indentation (typically 4 spaces) for code blocks

✗ Using `=` Instead of `==`

Use `==` for comparison, `=` is for assignment

✗ Forgetting Parentheses

Complex conditions need proper parentheses for clarity

Conclusion

The If-Else statement is an important decision-making tool in Python. It allows a program to check conditions and perform different actions based on the result. By using If-Else, programmers can make programs logical, interactive, and efficient. Understanding this concept helps in building more advanced and intelligent programs in Python.



Thank You!

If-Else statements are the building blocks of program logic. Master this concept, and you'll be ready to tackle more advanced programming challenges!

